

Executive Summary

This evaluation compares three AI-generated reports reviewing the **main.py (v1)** Random Forest regression model. While all three reports correctly identify that the model performs poorly—explaining only **22.63%** of the data variance—they differ significantly in the depth of their technical analysis and their utility for practitioners.

Claude-Day_2 is the superior report, providing a sophisticated "dual-audience" breakdown that translates complex metrics into plain English for executives while offering rigorous statistical analysis for technical teams. It identifies critical flaws, such as the mixing of heterogeneous data scales (counts vs. percentages), which none of the other reports fully explain. **ChatGPT-Day_2** provides a solid, professional overview with a clear methodology but lacks deep diagnostic insight. **Copilot-Day_2** offers a high-level summary but is the least complete, explicitly omitting recommendations and relying more on general assumptions than specific data distribution analysis.

Scores by Report

Criterion	ChatGPT-Day_2	Claude-Day_2	Copilot-Day_2
Correctness	9	10	8
Completeness	7	10	6
Clarity	9	10	8
Depth of Analysis	6	10	5
Usefulness	7	10	4
Grounding in Evidence	9	10	8
Total Average	7.8	10.0	6.5

Report 1: ChatGPT-Day_2

- **Strengths:** Highly professional structure with clear sections for methodology and limitations. It provides a transparent list of "Missing Contextual Information," which is useful for identifying gaps in the evaluation process.
- **Weaknesses:** The analysis is somewhat descriptive rather than diagnostic. It identifies the high MAPE (291.81%) but doesn't explain *why* the data structure caused it.

- **Notes:** No hallucinations were detected; however, it missed the opportunity to analyze the uninformative nature of constant categorical features (geography, sex, age).

Report 2: Claude-Day_2

- **Strengths:** Exceptional depth. It calculates the **RMSE/MAE ratio (1.43)** to identify a fat-tailed error distribution and recognizes that geography, sex, and age are constants that add zero discriminating power. It also correctly identifies that the **MAPE is skewed** by mixing percentage-based metrics with large patient counts.
- **Weaknesses:** None identified within the provided scope.
- **Notes:** Extremely well-grounded; even its assumptions are explicitly labeled and used to provide "lower-bound" performance estimates.

Report 3: Copilot-Day_2

- **Strengths:** Good at contextualising the data within a historical timeframe, mentioning COVID-19 disruptions as a source of non-stationarity.
- **Weaknesses:** Least useful for decision-making because it explicitly defers recommendations. It offers general assumptions about why the MAPE is high without the rigorous evidence-based confirmation seen in the Claude report.
- **Notes:** Omitted the crucial "Recommendations" section, stating information was needed to determine if they were in scope.

Comparison Table

Feature	ChatGPT-Day_2	Claude-Day_2	Copilot-Day_2
Target Audience	Technical & Executive	Distinct Executive & Technical Sections	General
Metric Interpretation	Descriptive	Diagnostic (e.g., RMSE/MAE ratio analysis)	General Assumption
Data Analysis	Methodology summary	Full profile (Min, Max, Mean, Std Dev)	High-level overview
Feature Insight	Top 10 list	Identifies uninformative constant features	Temporal vs. Categorical signal

Recommendations	General	Specific (Stratified models, Walk-forward CV)	Omitted
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Ranking

1. **Claude-Day_2** (Top-Ranked)
2. **ChatGPT-Day_2**
3. **Copilot-Day_2**

Why Claude-Day_2 is best: It is the only report that performs an actual **audit of the dataset** rather than just the model. By identifying that the geography, sex, and age columns were constants and that the target variable mixed radically different units (percentages vs. counts), it provides the only path forward for a practitioner to actually fix the model. It moves beyond "what" the results are to "why" they happened.

Improvement Suggestions

- **ChatGPT-Day_2:** Enhance the "Depth of Analysis" by looking at the distributions of the underlying data. Specifically, analyze why certain features like rolling_6 outperformed others and suggest specific architectural changes (like separate models for different data scales).
- **Claude-Day_2:** Maintain the current standard. To further improve, it could include a visual representation of the "fat-tailed error distribution" mentioned in the technical section to make it more accessible.
- **Copilot-Day_2:** Focus on completeness. Do not omit the recommendations section; even a preliminary set of suggestions based on standard ML best practices (like hyperparameter tuning or cross-validation) would be more useful than an empty section. Expand the interpretation of metrics beyond general assumptions.